Using data integration to improve current distributions of weasels in Ohio

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Introduction

Weasels are a key part of mammal communities in North America but have been understudied in research. Systematic survey efforts are lacking due to challenges in detecting (1) and identifying (2) weasels. With recent reports of population declines and possible range shifts across North America (3,4), we seek to understand the current status of long-tailed weasel (*Neogale frenata*) and least weasel (*Mustela nivalis*) across Ohio using a novel modeling approach that incorporates various types of observations.

Objectives:

- Describe current-day suitability and potential ranges of long-tailed and least weasels in Ohio.
- Develop integrated species distribution models (iSDMs) to understand possible weasel distributions to inform future management and research.

Methods

1. Compiled all available occurrence records between 2015 and 2025 (Fig. 2) from the following sources:
   - **Presence-only data:** museum specimens (GBIF), and citizen-science observations from iNaturalist and records reported to the Ohio DOW.
   - **Presence-absence data:** Camera trap and AHDriFT (Adapted Hunt Drift Fence Technique) data collected by authors and publicly available images on Wildlife Insights.
2. Created iSDMs using integrated nested Laplace approximation (INLA) via the *intSDM* package (5) in R with various combinations of covariates (6).
   - Uninformative prior distribution set for each model.
   - Triangulation mesh generated for spatial correlation calculations.
   - **Environmental Covariates:** snow-water equivalent (SWE) and annual maximum temperature (tmax) downloaded from Daymet (7).
   - **Land Cover Covariates:** percent cover of grassland, agriculture, developed land, and water derived from the National Land Cover Database (8).

Figure 1. Long-tailed weasel and short-tailed weasel. Image credit: iNaturalist @smays and @a_ma_bird.

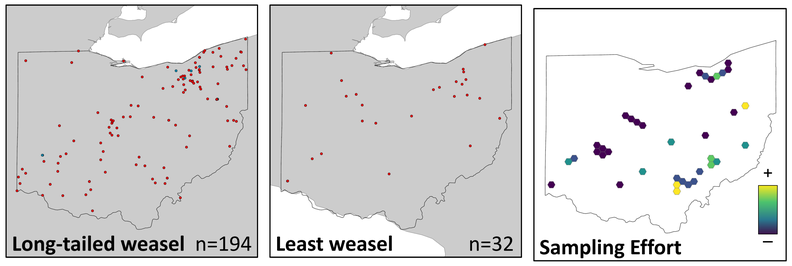

Figure 2. Compiled weasel occurrences and sampling effort in Ohio. IUCN species' ranges are shaded in gray. Red points for occurrences indicate presence-only observations and blue points indicate presence-absence observations. Sampling effort calculated using total number of days cameras were active, summarized in 50 km^2 cells (max: 4,676; min: 13).

Results

We collected 159 presence-only occurrences and 32 presence-absence occurrences (Fig. 2; Table 1). Camera trap and AHDriFT sampling totaled 31,044 camera-nights (Fig. 2).

Predicted suitability for long-tailed weasels was highest in the northeastern and southwestern portions of the state and showed relatively smooth, continuous distribution (Fig. 3). In contrast, least weasel suitability was highest in the northeast, and in and around grasslands of the southeast, with a more patchy distribution across the landscape. Long-tailed weasels showed lower predicted suitability in the northwest and a small portion of the southeast, while least weasels had lower predicted suitability across much of the southern portion of the state.

Long-tailed weasel suitability was negatively associated with agriculture, water cover, and maximum temperature. Least weasel suitability showed negative relationship with maximum temperature and positive relationship with developed land and grasslands (Fig 4).

Table 1. Statistics for models of each species. # PO and PA are the number of data points per data type. RSS is residual sum of squares.

species	# PO	# PA	Log marginal likelihood	RSS
Long-tailed	127	67	-781.957	937.737
Least	32	0	-125.719	609.078

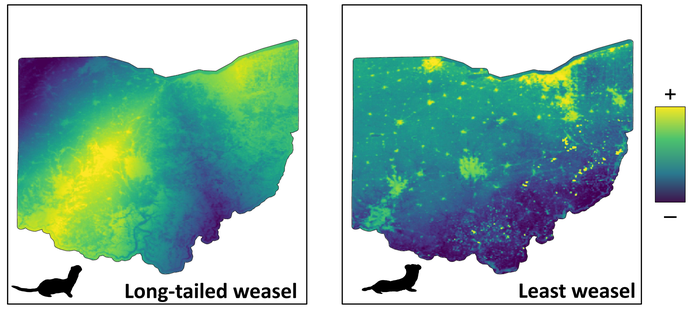

Figure 3. iSDM suitability maps for long-tailed and least weasels. Color scale represents mean predicted suitability.

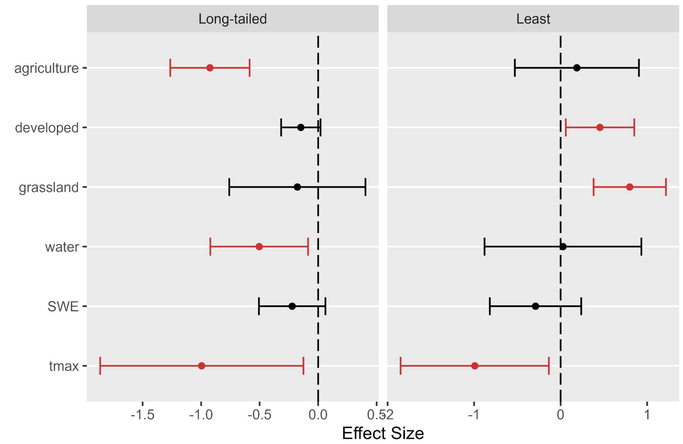

Figure 4. Effect sizes of the covariates included in the model for each species. Error bars represent credible intervals. Significant covariates are indicated in red.

Discussion

Both weasel species show higher predicted suitability in the northeastern portion of the state, but diverge in other areas. The species appear to prefer different habitat types; long-tailed weasels are negatively associated with agricultural land cover, and least weasels show a clear association with grasslands and developed (urban) environments (Fig. 4). These results are generally consistent with previous modeling efforts which found the highest suitability for both species in the northeast and similar patches of higher suitability across other portions of the state (3).

The negative relationship between weasel suitability and maximum temperature suggests the potential for future range shifts, as both species may be sensitive to increasing temperatures (Fig. 4). In contrast, neither species showed a significant relationship with snow water equivalent (SWE), indicating that snow may not be a strong limiting factor for weasel distribution in Ohio.

Future monitoring should prioritize areas with high suitability that remain unsampled and areas that have been historically underrepresented (e.g. northwestern Ohio and select grassland habitats). Targeting both will improve statewide coverage, and facilitate model validation. Future modeling efforts should seek to refine these results by including additional data from Ohio and neighboring states, additional environmental covariates, and refined prior selection.

Acknowledgments

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References

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